

supplementary test 3

Total points 12/60 ?

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Hyunseo



✗ 1 *

0/2

2. During the 1850s, there was an outbreak of cholera in London. A doctor, John Snow, suspected that the source of the disease was the water that people used. On a map he marked the homes of people with cholera, and the location of the water pumps in the area, as shown.

Which pump was likely to be the source of the cholera outbreak?

☒ A

✗

☐ B☐ C☐ D

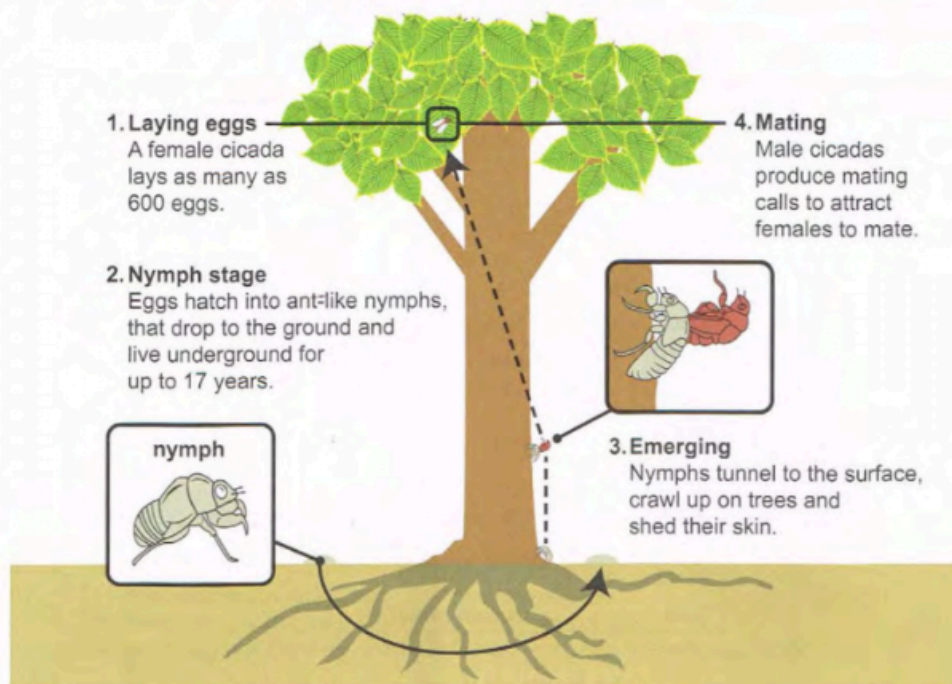
Correct answer

☒ B

X 2

0/2

5. The diagram shows the life cycle of a cicada.



Cicadas are prey for birds, squirrels and cicada killer wasps.

Which of the following helps the species to survive?

- (A) having a hard shell to protect themselves
- (B) having strong front legs to fight off predators
- (C) laying lots of eggs to outnumber predators
- (D) changing body form to confuse predators

☒ A

☐ B

☐ C

☐ D

Correct answer

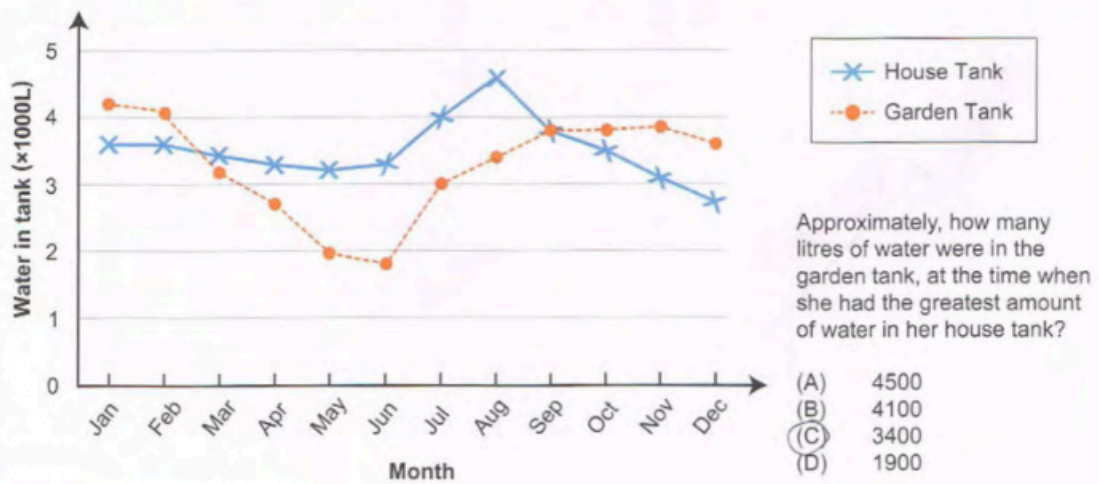
☒ C



✗ 3

0/2

6. Karen has two water tanks, one for her house and another for her garden. She measured and recorded the amount of water in each tank at the end of each month over a year and summarised her results in this graph.


☒ A

✗

☐ B

☐ C

☐ D

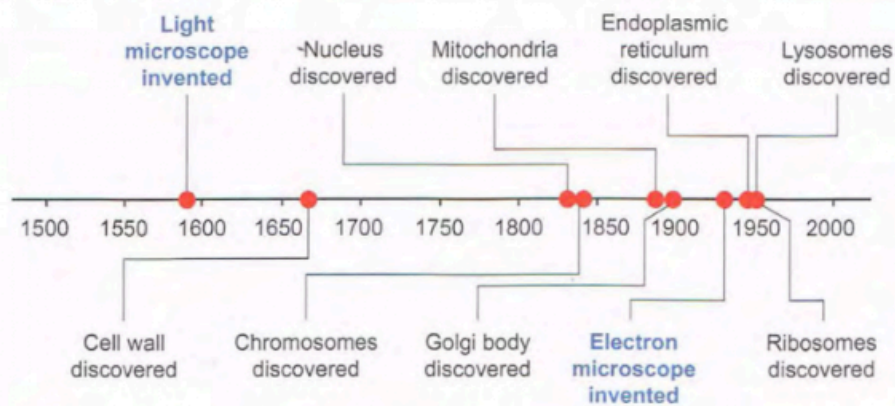
Correct answer

☒ C


✓ 4 *

2/2

7. The timeline shows the years when two different types of microscopes were invented and the years when different parts of cells were discovered.



Which parts of cells can be seen using a light microscope?

- (A) Cell wall, chromosomes, mitochondria
- (B) Cell wall, chromosomes, lysosomes
- (C) Chromosomes, endoplasmic reticulum, nucleus
- (D) Chromosomes, nucleus, ribosomes

☒ A☐ B☐ C☐ D

✖ 5

0/2

8. Symbiosis is the name for a close relationship between two different species.

The table below compares different types of symbiotic relationships.

Relationship	Description
Amensalism	One organism is harmed but there is no effect on the other.
Commensalism	One organism benefits but the other is not affected.
Mutualism	Both organisms benefit.
Parasitism	One organism benefits at the expense of the other.

When Galapagos tortoises rise up on their legs, Galapagos finches can eat parasites from underneath the tortoise's body.

What type of symbiosis is this relationship between the tortoise and the finch?

- (A) Amensalism
- (B) Commensalism
- (C) Mutualism
- (D) Parasitism



☒ A

✖

☐ B

☐ C

☐ D

Correct answer

☒ C



✖ 6

0/2

For questions 9 and 10 use the information below.

Steel is made of iron and small quantities of other elements.

The table summarises the proportion by mass, of elements other than iron present in four different types of steel.

Steel No.	Other element present (%)									
	C	Si	Mn	P	S	Cr	Ni	Mo	V	N
1	0.96	1.50	0.80	≤ 0.025	≤ 0.020	1.0	≤ 0.10	≤ 0.07	≤ 0.03	≤ 0.03
2	1.50	1.50	0.80	≤ 0.025	≤ 0.020	1.0	≤ 0.10	≤ 0.07	2.0	≤ 0.03
3	2.00	1.50	0.80	≤ 0.025	≤ 0.020	1.0	≤ 0.10	≤ 0.07	4.0	≤ 0.03
4	0.95	1.28	0.84	0.007	0.005	1.23	0.14	0.03	0.09	0.011

9. What is the approximate percentage of iron present in Steel 4?

- (A) less than 5% (B) between 93% and 95%
(C) more than 97% (D) between 95% and 97%

☒ A

✖

☐ B

☐ C

☐ D

Correct answer

☒ D



✓ 7

2/2

10. To convert Steel 1 to Steel 2, the percentage of carbon (C) and vanadium (V) need to be increased.

How would the manufacturers of the metals allow for this?

- | | | | |
|-----|--|-----|--|
| (A) | reduce the percentage of Fe | (B) | increase the percentage of Fe |
| (C) | reduce the percentage of Si, Mn and Cr | (D) | reduce the percentage of P, S, Ni and Mo |

☒ A



☐ B

☐ C

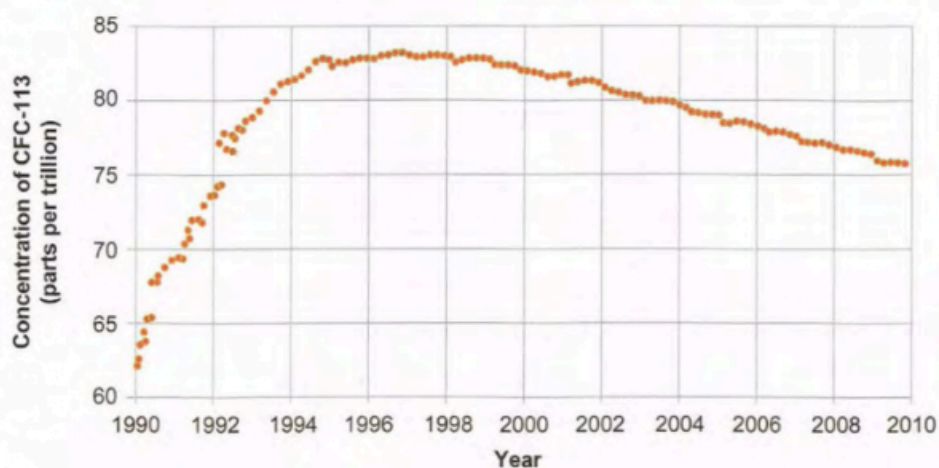
☐ D



✗ 8

0/2

12. Chlorofluorocarbons (CFCs) are a class of chemicals that cause the holes in the atmosphere's ozone layer. The manufacture of these chemicals mostly stopped on January 1, 1996. The graph represents the concentration of one CFC (CFC-113) at one location in the atmosphere during the years 1990 to 2010.



The concentration of CFC-113 took about 13 years to fall from its maximum level, to 76 parts per trillion.

Approximately how long had it taken for the concentration of CFC-113 to rise from 77 parts per trillion to its maximum level?

- (A) 3 years (B) 5 years (C) 7 years (D) 13 years

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ B

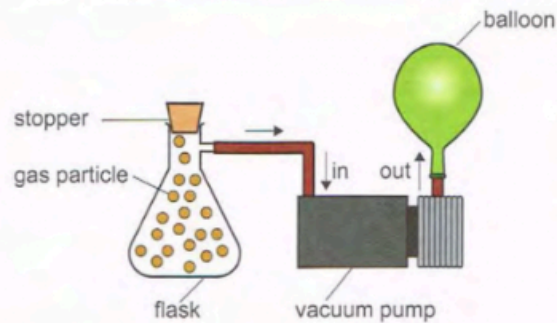


X 9

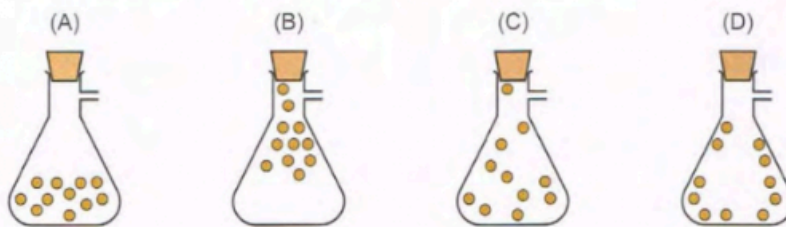
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14. Scientists use the Particle Theory of Matter to explain the behaviour of gases. According to this model, the particles that make up a gas always spread throughout any container.

Mr. Jones set up a vacuum pump, a glass flask and a balloon. He used the vacuum pump to remove some gas from the flask to inflate the balloon.



Which diagram best represents the gas particles that remain in the flask?

☒ A

X

☐ B☐ C☐ D

Correct answer

☒ C

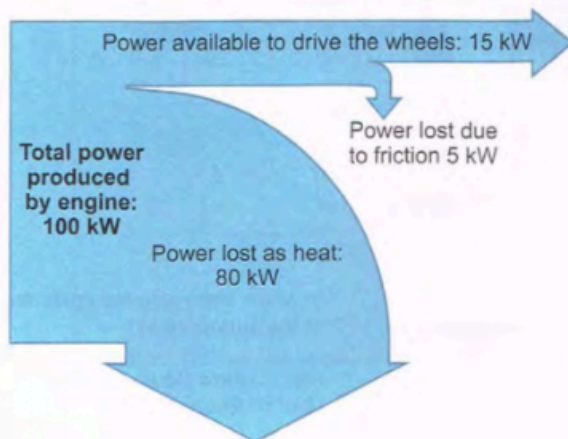
✗ 10

0/2

15. When we use machines to convert energy from one form into another, they produce some unwanted waste energy. The efficiency of a machine can be calculated using the equation

$$\text{efficiency} = \frac{\text{available power output}}{\text{total power produced}} \times 100\%$$

This diagram represents the forms of power produced in a car.



What is the efficiency of the car?

- (A) 75 %
- (B) 20 %
- (C) 15 %
- (D) 10 %

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ C



X 11

0/2

16. The table lists the densities of some common liquids. A liquid with a lower density will float on top of a liquid with a higher density.

Material	Density (g/cm ³)
water	1.00
vegetable oil	0.91
detergent	1.03
glycerin	1.26
honey	1.36
baby oil	0.82

Ben carefully poured some of these liquids into a cup. He poured the denser ones in before the less dense ones so that they did not mix. He labelled the water and the glycerin, but not the other layers.



Which of these is one way that the liquids could have been arranged?

	1	2	3
(A)	baby oil	vegetable oil	honey
(B)	baby oil	detergent	honey
(C)	vegetable oil	honey	detergent
(D)	honey	detergent	baby oil

☒ A

X

☐ B

☐ C

☐ D

Correct answer

☒ B


✓ 12

2/2

17. All the measurements we make contain an inaccuracy called an error. The table lists some sources of error.

Error	Source
Lag time	The time an instrument takes to stabilise and give the correct reading.
Limit of reading	An instrument cannot be used to measure a quantity smaller than half the smallest division on the scale.
Parallax	The scale is being read at an incorrect angle.
Zero error	The instrument does not show a zero reading when nothing is being measured.

Two students used the same thermometer to measure the temperature of boiling water in a container. The first student measured the temperature of the boiling water to be 88 °C. The second student measured it to be 100 °C.

What was the source of error in the first student's reading?

(A) lag time
(C) parallax

(B) limit of reading
(D) zero error

☒ A



☐ B

☐ C

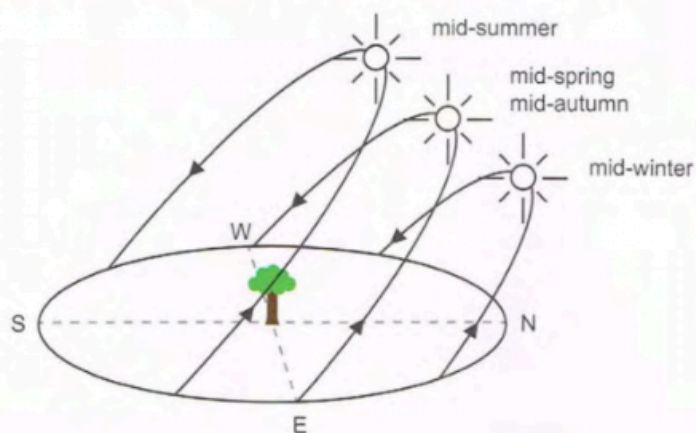
☐ D



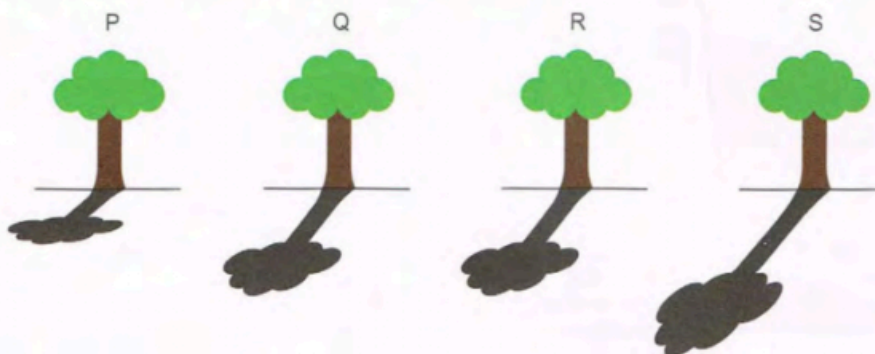
✓ 13

2/2

18. This diagram shows the Sun's path across the sky in the southern hemisphere at different times of the year.



Tina drew diagrams to record the position of the shadow formed by a tree at noon in mid-summer, mid-autumn, mid-winter and mid-spring.



Which of these matches each drawing with the season in which Tina drew it?

	Diagram P	Diagram Q	Diagram R	Diagram S
(A)	mid-summer	mid-spring	mid-autumn	mid-winter
(B)	mid-summer	mid-autumn	mid-winter	mid-spring
(C)	mid-autumn	mid-spring	mid-summer	mid-winter
(D)	mid-winter	mid-autumn	mid-spring	mid-summer

☒ A

☐ B

☐ C

☐ D


✖ 14

0/2

19. The table below describes four methods which speed up a chemical reaction.

Method	How it works
Add a catalyst	A catalyst provides an easier reaction pathway, so more of the particles will react.
Increase concentration	The number of particles per unit volume is increased so particles from the different chemicals are more likely to come into contact.
Increase surface area	The chemicals can only react at the surface where they come in contact with each other. Making the particles smaller increases the total surface area.
Increase temperature	The particles that make up the chemicals involved in the reaction move faster, so they are more likely to come into contact with each other.

Eric's science teacher was demonstrating the reaction between sulfuric acid and sugar.

In beaker X she mixed the acid with normal sugar crystals. In beaker Y she used sugar that she had ground up into a powder.

In which container should the acid and sugar have reacted faster and what is the reason for this?

	The reaction was faster in	Reason
(A)	beaker X.	The sulfuric acid is a catalyst.
(B)	beaker X.	The normal sugar crystals have a higher concentration.
(C)	beaker Y.	The particles in the powdered sugar move faster.
(D)	beaker Y.	The sugar powder has more surface contact with the acid.

☒ A

✖

☐ B☐ C☐ D

Correct answer

☒ D

✗ 15

0/2

20. The table lists some information about five acids.

Name	Formula	Boiling point (°C)	Heat of vaporisation (kJ/mol)	Flash point (°C)	Density (g/cm ³)
Formic acid	HCOOH	100.6	46	68	1.214
Acetic acid	CH ₃ COOH	117.9	49	40	1.044
Propionic acid	C ₂ H ₅ COOH	140.8	57	51	0.988
Butyric acid	C ₃ H ₇ COOH	163.3	64	76	0.952
Valeric acid	C ₄ H ₉ COOH	185.5	70	88	0.935

Which property of the acids does **NOT** follow a trend?

- (A) Boiling point
- (B) Heat of vaporisation
- (C) Flash point
- (D) Density

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ C

✗ 16

0/2

21. Jane carried out an investigation to determine the best conditions under which tomato plants grow. She compared the plants by counting how many leaves each plant had.

She grew four groups of three tomato plants, each under different conditions. The table shows her results.

Conditions	Plant height (cm)			Number of leaves		
	Plant			Plant		
	1	2	3	1	2	3
Full sun, outside	85	74	81	16	14	15
Partial shade, outside	73	92	78	9	12	11
Full shade, outside	40	43	34	5	6	4
Partial shade, indoors	55	49	52	7	8	8

What could Jane do to improve her investigation?

- (A) use different types of plants
- (B) use larger groups of tomato plants
- (C) keep all the plants outside in the sun
- (D) give the plants twice as much water each day

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ B

X 17

0/2

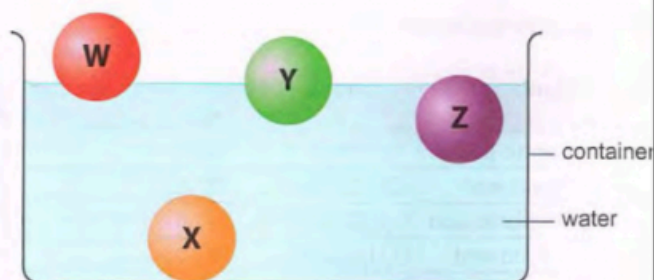
22. Objects that have a density greater than that of water sink in water. Objects that have a density less than that of water float on water. The lower the density of the material, the higher it will float.

Sam had four solid spheres, each one made from a different type of wood. She placed the wooden spheres into a container with water.

The table shows the density of each type of wood. The density of water is 1.0 g/cm^3 .

The diagram shows what she observed.

Type of wood	Density (g/cm^3)
balsa	0.14
box	1.10
gum	0.95
maple	0.53



Which of the following correctly matches a sphere with the wood from which it is made?

	W is made of	Z is made of
(A)	balsa	maple
(B)	balsa	gum
(C)	box	maple
(D)	box	gum

☒ A

X

☐ B

☐ C

☐ D

Correct answer

☒ B


✗ 18

0/2

23. Research has shown that organically grown food plants retain more minerals than those grown using non-organic methods. The table lists the mineral content of three food plants grown organically and non-organically.

Food	Mineral content (milliequivalents)					
	Organically grown food			Non-organically grown food		
	Calcium	Magnesium	Potassium	Calcium	Magnesium	Potassium
Cabbage	60.0	43.6	148.3	17.5	15.6	53.7
Lettuce	71.0	49.3	176.5	16.0	13.1	53.7
Tomatoes	23.0	59.2	148.3	4.5	4.5	58.6

Based on the information in the table, which of the following statements is correct?

- (A) The ratio of the potassium content to magnesium content is the lowest in organically grown cabbage.
 (B) Mineral content in organically grown food is not always higher than that in non-organically grown foods.
 (C) The ratio of the potassium content of organic to non-organic vegetables is the highest in lettuce.
 (D) The calcium content in organically grown tomatoes is the lowest of the three minerals in all vegetables.

☒ A

✗

☐ B

☐ C

☐ D

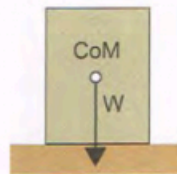
Correct answer

☒ C


✗ 19

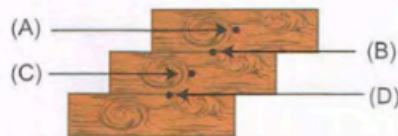
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25. Every object has a centre of mass (CoM). It is the point where all its mass appears to be concentrated and where its weight force (W) acts. If an object has a regular shape, its CoM lies at its geometrical centre.



A number of identical wooden blocks are stacked up on a flat surface, and glued together as shown in the diagram.

Which point is the CoM of the stack of blocks?

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ C

✗ 20

0/2

For questions 26 and 27 use the information below.

A group of students wanted to find out if the time it took for water to freeze depends on the volume of the water. Part of their report is shown below.

The effect of volume on freezing time

Aim: To determine if the volume of water affects how long it takes to freeze

Equipment: 250 mL measuring jug, 4 plastic cups, water, freezer, timer

Method:

1. Measure 50 mL of water in the measuring jug and transfer the water into a plastic cup.
2. Repeat step 1 so that the 2nd cup contains 100 mL, the 3rd cup contains 150 mL and the 4th cup contains 200 mL.
3. Place all four cups in the freezer.
4. Check each cup every 10 minutes and record the time when the water freezes.

26. What must the students do to make their procedure a fair test?

- (A) use a measuring cylinder to measure the volume of water
- (B) keep the volume of water constant every time they do the experiment
- (C) conduct the experiment five times to see if the results are the same
- (D) make sure the starting temperature of each cup of water is the same

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ D



✖ 21

0/2

27. Which of the following most significantly improves the accuracy of the test?

- (A) use a stopwatch to measure the time
- (B) use a measuring cylinder instead of a measuring jug to measure the volume of the water
- (C) check the cups in the freezer every 2 minutes instead of every 10 minutes
- (D) conduct the experiment for each cup five times and calculate the average of the results

☒ A

✖

☐ B☐ C☐ D

Correct answer

☒ C

X 22

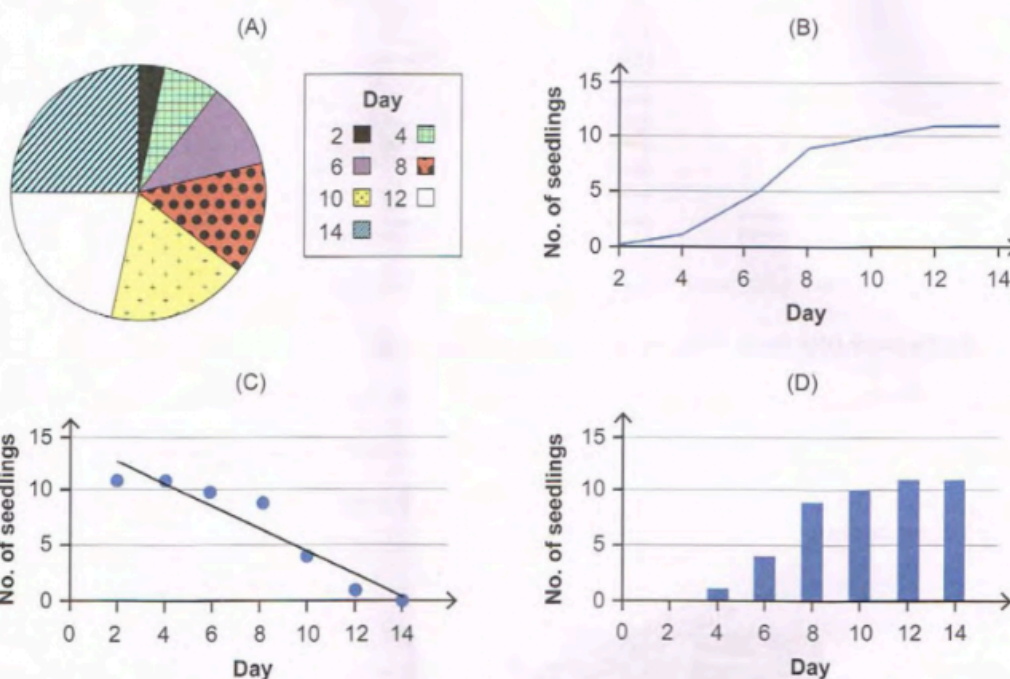
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30. Graphs are used to present data. The type of data to be presented determines what type of graph will be the most appropriate.

Sonya's class planted some tomato seeds as part of an investigation. Every two days, they counted the number of seeds that had germinated. Their results are listed in the table.

Day	2	4	6	8	10	12	14
Number of seedlings	0	1	4	9	10	11	11

Which graph is the most appropriate to present the students' data?


☒ A

X

☐ B

☐ C

☐ D

Correct answer

☒ D


✖ 23

0/2

31. Full fat milk contains about 3.5% fat by mass. Cream contains 35% fat by mass.

Kanye mixed 400 grams (g) of full fat milk with 200 g of cream.

How many grams of fat were there in 150 g of his mixture?

(A) 14

(B) 21

(C) 42

(D) 84

☒ A

✖

☐ B

☐ C

☐ D

Correct answer

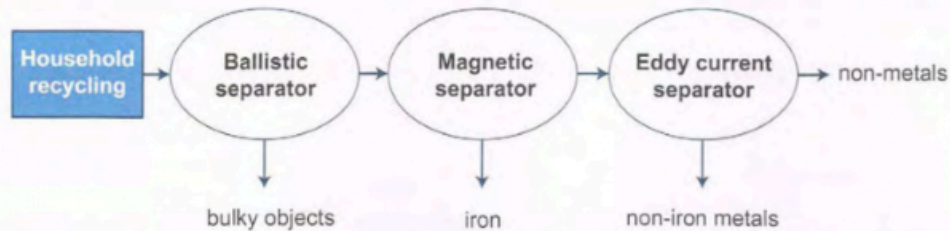
☒ B



✓ 24

2/2

34. Household waste that can be recycled contains different types of materials. The materials can be separated using their properties. The flowchart summarises one process used for separating recyclable waste materials.



In which of the following sets of recyclable waste can the components be separated completely using this process?

- (A) newspaper, iron nails, copper wire, large cardboard box
- (B) aluminium can, iron nails, copper wire, large cardboard box
- (C) newspaper, aluminium can, copper wire, large cardboard box
- (D) plastic bucket, iron nails, copper wire, large cardboard box

☒ A☐ B☐ C☐ D

✗ 25

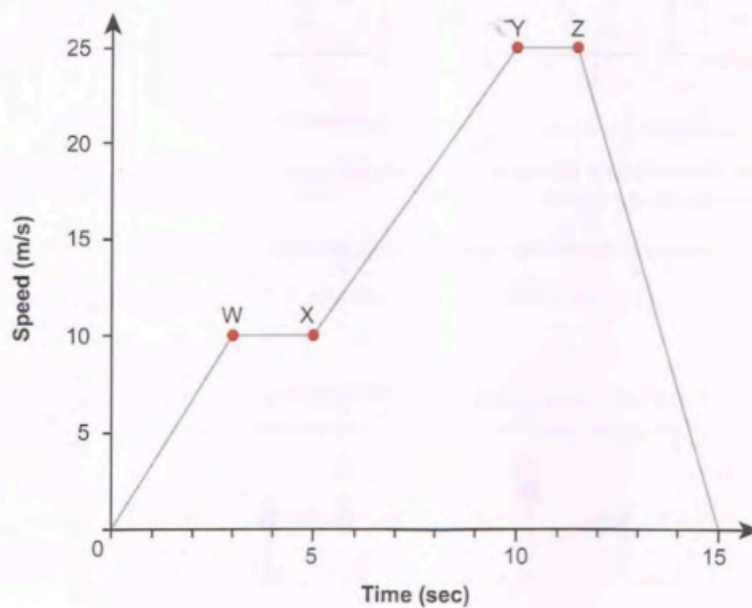
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For questions 38 and 39 use the information below.

The speed of an object can be calculated using the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Chris was driving her car along a straight road. The graph represents the speed at which her car was moving as time went on.



38. How far did the car travel from W to X in metres?

- (A) 0 (B) 2 (C) 10 (D) 20

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ D



✖ 26

0/2

39. Which statement about the car's motion matches the information in the graph?

- (A) Point Y represents the time at which the car reached its greatest distance from the starting point.
- (B) Point Z represents the time at which the car reached its greatest speed.
- (C) After 15 seconds the car was back where it started from.
- (D) The speed of the car changed fastest in the last three seconds.

☒ A

✖

☐ B☐ C☐ D

Correct answer

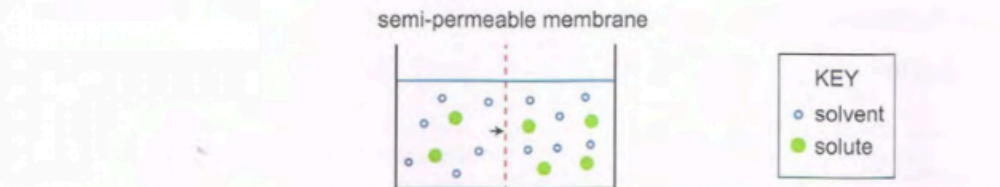
☒ D

✗ 27

0/2

40. Osmosis is the movement of solvent molecules (e.g. water) through a semi-permeable membrane from a region of lower solute (e.g. sugar) concentration to that of a higher solute concentration until the concentrations of both sides are equal. The semi-permeable membrane allows the solvent, but not the solute, to move across.

A tank is divided by a semi-permeable membrane and partly filled with two sugar solutions of different concentration, as shown.



Which diagram shows the results after some time?

☒ A

✗

☐ B☐ C☐ D

Correct answer

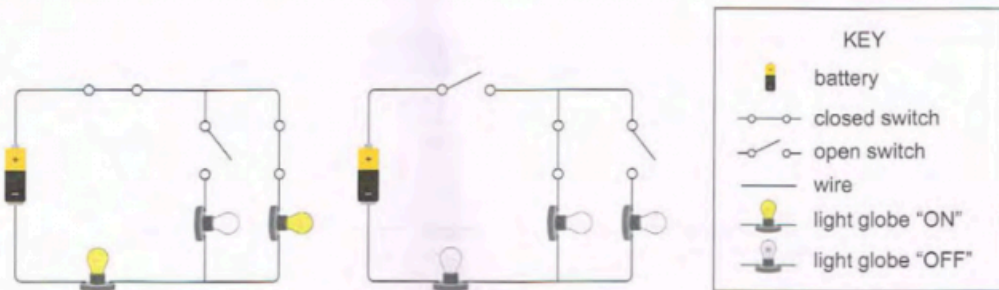
☒ D

✗ 28

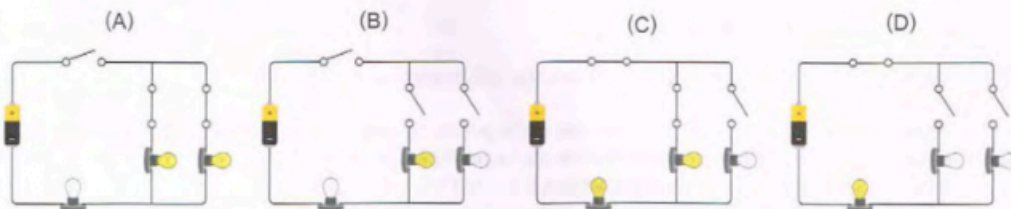
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41. Electricity can flow when it has a source of energy like a battery to push it and an unbroken path to follow. This unbroken path is called a circuit.

The following circuit diagrams show how switches affect light globes in a circuit.



Which circuit diagram is correct?


☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ C

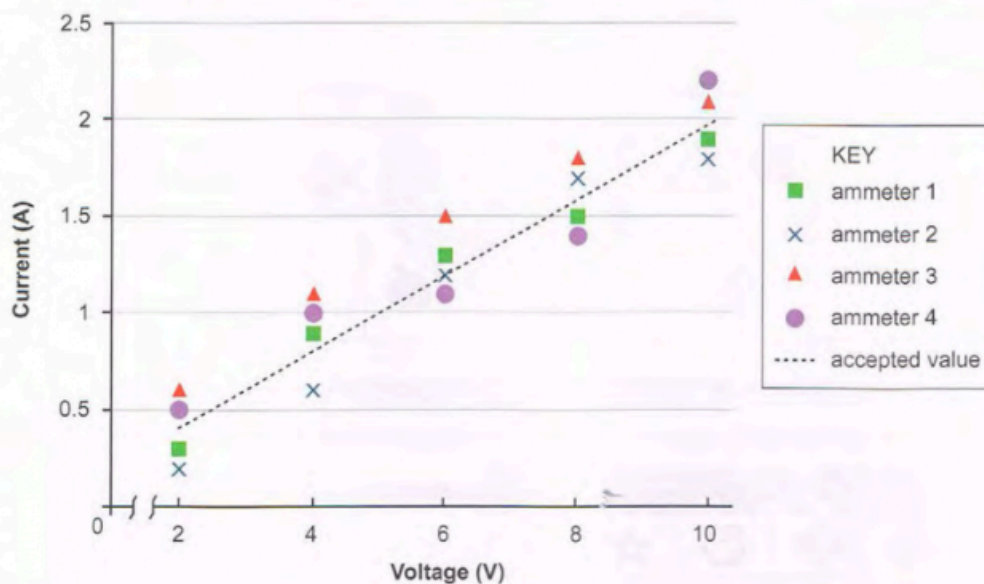

✓ 29

2/2

For questions 42 and 43 use the information below.

Accuracy refers to the closeness of a measurement to an accepted value. Precision refers to the closeness of two or more repeated measurements under the same conditions.

Four different ammeters were used to measure the electric current through the same resistor as the voltage was increased. The results are graphed below.



42. Which ammeter gave the most accurate results?

- (A) ammeter 1 (B) ammeter 2 (C) ammeter 3 (D) ammeter 4

☒ A

☐ B

☐ C

☐ D



✖ 30

0/2

43. Which of the following statements about the precision of the ammeters is correct?

- (A) Ammeter 1 is the most precise.
- (B) Ammeter 3 is the most precise.
- (C) We cannot conclude because the measurements varied too much from each other.
- (D) We cannot conclude because all measurements were made at different voltages.

☒ A

✖

☐ B☐ C☐ D

Correct answer

☒ D

*

☒ Option 1

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